Real Time Network Intrusion Detection using PySpark and Machine Learning

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***Abstract*— The increase in both the frequency and complexity of cyber threats has made network security more difficult to manage. This research tackles the use of PySpark and Machine Learning technologies for improving the efficiency of Network Intrusion Detection Systems (NIDS) in responding to different forms of network attacks. Utilizing PySpark's distributed computing resources allows for large scale data processing and model training, which tackles the concerns of current-day cyber challenges involving massive amounts of data. After thorough testing, it was proven that the Naive Bayes classifier performed the best with 85% accuracy and with no indications of overfitting, thus confirming dependable and solid performance in practical applications.**

Keywords— Network Intrusion Detection, PySpark, Machine Learning, Cybersecurity

I. INTRODUCTION

In today's world, the intertwined nature of life has rendered cyber-security a vital concern. Breaches pertaining to data, phishing, ransomware as well as advanced persistent threats: each constitute significant risks towards the security of the information. Not only individuals, but organizations too are becoming more and more dependent on the digital medium for communication as well as storage purposes. This gradual shift has resulted in an increased cyber-attack potential which poses dire threats pertaining to financial, reputation, and operational factors.

With regard to these concerns, network beam intrusions stands out as the most common and pernicious attack. Sophisticated attacks, such as Denial of Service (DoS), Remote to Local (R2L), User to Root (U2R), and Probe form an attack vector by taking advantage of a network's security loophole out of its stealth scope. These can range from theft of sensitive information to storage of data and services denial: causing catastrophic interruptions and fundamentally tampering with essential services.

DoS attacks are designed to overwhelm networks or systems, making them inaccessible to legitimate users. These attacks operate by flooding the target with massive volumes of traffic, consuming critical resources including bandwidth, CPU, and memory. This resource exhaustion leads to service degradation or complete system shutdown, effectively disrupting normal operations. By exploiting infrastructure vulnerabilities such as unprotected servers or weak protocols, attackers can successfully cause widespread service outages. The impact on businesses can be severe, often resulting in significant downtime, substantial revenue losses, and long-term reputational damage.

R2L attacks occur when attackers exploit vulnerabilities in remote systems to gain unauthorized access to local machines within a network. These attacks typically begin with the attacker establishing a connection from an external system to a remote network, utilizing techniques such as password guessing, software vulnerability exploitation, or malicious code injection. Upon successfully gaining access, attackers can escalate their privileges, enabling them to control local resources, exfiltrate data, or execute malicious actions within the compromised network. The particular danger of R2L attacks lies in their ability to bypass perimeter defenses, allowing direct access to internal network assets.

U2R attacks are a type of privilege escalation attack where an attacker gains initial access to a system as a regular user and then escalates their privileges to root or administrative level. This type of attack typically exploits vulnerabilities in the system's software, such as buffer overflows or misconfigured permissions, to execute unauthorized commands or gain elevated access. Once the attacker achieves root access, they can compromise the entire system, install malicious software, alter configurations, and access sensitive data. U2R attacks are highly dangerous because they allow the attacker to take full control of the compromised system, potentially affecting other connected systems in the network.

Probe attacks consist of scanning and investigating a network to collect details about its structure, vulnerabilities, and potential weaknesses. In these attacks, the attacker typically uses automated tools to discover open ports, services, and security flaws in the target system. While probing does not immediately cause damage, it serves as a precursor to more harmful attacks, such as DoS, R2L, or U2R. By gaining insight into the network's defenses, the attacker can plan

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subsequent attacks with greater precision. Probe attacks are often difficult to detect because they are usually low-profile activities, but they provide critical intelligence that can be used in a larger, more devastating cyberattack.

Machine learning and PySpark create an effective solution for Network Intrusion Detection (NID) by combining advanced threat detection capabilities with scalable data processing. Through pattern recognition and continuous learning, ML models can identify and adapt to diverse attack types, while becoming more resilient through regular retraining. PySpark complements this adaptability with its distributed architecture, enabling efficient processing of large-scale network data in real-time.

II. Literature Survey

The authors performed an evaluative assessment of machine learning algorithms to provide approaches for open-source, open-data network intrusion detection (NID) and tested three methods: Naive Bayes, Support Vector Machine (SVM), and Random Forest classifiers. The results indicated that Random Forest performed best in detecting known attack patterns, a result that the authors asserted was made possible because of the ensemble learning framework used that is helpful for multiple intrusion signatures. The authors provided this study as a validation of machine learning systems as effective ways to improve IDS accuracy for known attack types; however, the study had notable limitations. The study did not include an evaluation of detection in real-time, nor did the authors comment on potential overfitting, simply noting that the performance was exceptionally high with the training data. The models are used to demonstrate the success of newly proposed learning systems, but as the results were as high as they were and no mention of real-time use were reported, it was unclear whether these models would perform well in real deployment. [1]

Lirim Ashiku1 and Cihan Dagli researched and implemented a new approach to deep learning for network intrusion detection (NID) that used a Convolutional Neural Network (CNN) that would be trained on their data. Their overall architecture comprised of convolutional layers that had been intentionally designed and enhanced with regularization and optimized hyperparameters. While the authors are able to claim they had achieved high accuracy and top performance against other models, they did identify difficulty in detecting attack classes that did not appear as frequently. This demonstrated a typical and much-discussed aspect of intrusion detection systems - class imbalance. [2]

Sivamohan and Sridhar integrated bidirectional Long Short-Term Memory (BiLSTM) networks with explainable AI (XAI) methods to create a NID framework, the has achieved great success with an accuracy of 98.2%. However, such great success comes with the issue of overfitting the model. [3]

Ejbali and Mohamed fused SARSA based reinforcement learning with deep neural networks to build a hybrid model for NID that across all metrices, including accuracy, F1-score, precision along with recall, outperformed traditional techniques like CNN, DNN and SVM. The hybrid model does perform better; however, it suffers from class imbalance in more intricate network topologies. [4]

The authors have developed a sophisticated multi-stage optimized machine learning framework for NID that addresses key challenges in computational efficiency and detection accuracy. Their research offers several key contributions, including examining the effect of oversampling methods on training sample size, comparing information gain and correlation-based feature selection approaches, and exploring various ML hyperparameter optimization techniques. [5]

Yuansheng Dong, Rong Wang, and Juan He have introduced a real-time NIDS that leverages the combined power of big data processing, natural language processing, and deep learning technologies to address traditional limitations. Their innovative approach integrates Flume for real-time log collection and Flink for stream processing, while tackling high-dimensional data challenges through a novel self-encoder-based dimension reduction method. The core of their system, the AE-AlexNet model, combines Auto-Encoder with AlexNet neural network architecture. [6]

The authors have presented a comprehensive study on NIDS utilizing multiple machine learning approaches. Their research focused on both signature-based and anomaly-based detection methods, implementing SVM, Naïve Bayes, and Artificial Neural Network algorithms. The study's significance lies in its evaluation of these algorithms using the widely-recognized KDDCUP99 and NLS-KDD benchmark datasets, providing valuable insights into their effectiveness in detecting malicious activities in real-time network environments. Their experimental analysis demonstrated the comparative performance of these algorithms in identifying unauthorized access and intrusion attempts, contributing to the advancement of computer network and virtual machine security frameworks. [7]

The authors conducted an experiment of real-time NID based on deep learning, using a cloud-based prototype system to compare binomial and multinomial classification models. They compared deep learning models built with H2O and DeepLearning4J libraries with traditional machine learning approaches. While their H2O-based deep learning models performed with high accuracy rates of over 99.5% on the NSL-KDD dataset, a significant limitation was faced in the form of overfitting, as evident from the dramatic drop in performance to 83% accuracy when tested on the independent test dataset. This staggering difference between training and test performance means that their models, despite having high training accuracy, were unable to generalize to new, unseen network traffic patterns.[8]

The authors suggested a technique to NIDS by implementing Self-taught Learning, a deep learning technique, to address the challenges of detecting unanticipated and unpredictable network attacks. Their research employed the NSL-KDD benchmark dataset to validate the performance of their suggested system. [9]

Taehoon Kim and Wooguil Pak created a novel real-time NID with a hybrid of session-based and packet-based classification methods. Their work suggested a hybrid

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classifier that utilizes a deferred decision model with cumulative features for single packet integration. The merit of their work lies in addressing the significant problem of detection delay without sacrificing high accuracy—a common trade-off in traditional NIDS designs. Their hybrid architecture demonstrated excellent performance by achieving a smooth balance between real-time detection ability and accuracy, along with high scalability to support environments with high traffic rates and concurrent sessions. [10]

The examined studies demonstrate improvements in network intrusion detection, but they also highlight significant drawbacks, such as overfitting, a lack of real-time assessment, difficulties with class imbalance, and underrepresented attack types. Many approaches struggle to generalize to unseen data, despite achieving high accuracy, which raises questions about their practical implementation. Their applicability is further limited by problems like scalability and computational efficiency. Our strategy aims to get past these obstacles.

III. PROPOSED METHODOLOGY

A. Dataset

In this study, network traffic is analyzed and classified for intrusion detection using the NSL-KDD dataset. About 25,000 labeled instances are included, representing both typical traffic and different kinds of network attacks, such as Denial of Service (DoS), Remote to Local (R2L), User to Root (U2R), and Probe attacks. In addition to a label indicating whether the traffic is normal or an attack, each instance includes 41 features covering content-based details, time-based traffic statistics, and basic network attributes. By providing a more balanced distribution of records, this dataset tackles the problem of class imbalance and is therefore ideal for machine learning model training and evaluation. The distribution of various attack types and a comparison of instances of normal and abnormal traffic are shown in Figures 1 and 2.

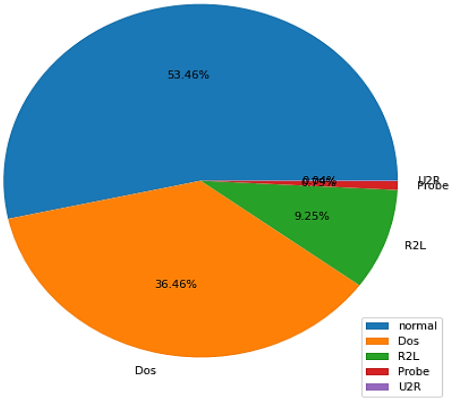


Fig 1. Distribution of different attack types in the dataset

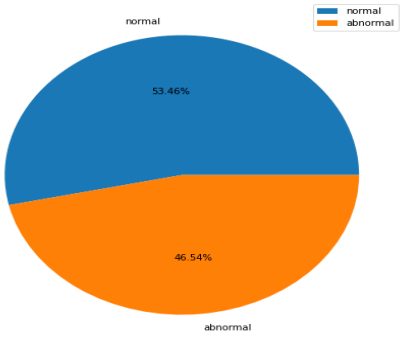


Fig 2. Distribution of normal and abnormal labels in the dataset

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B. Architecture

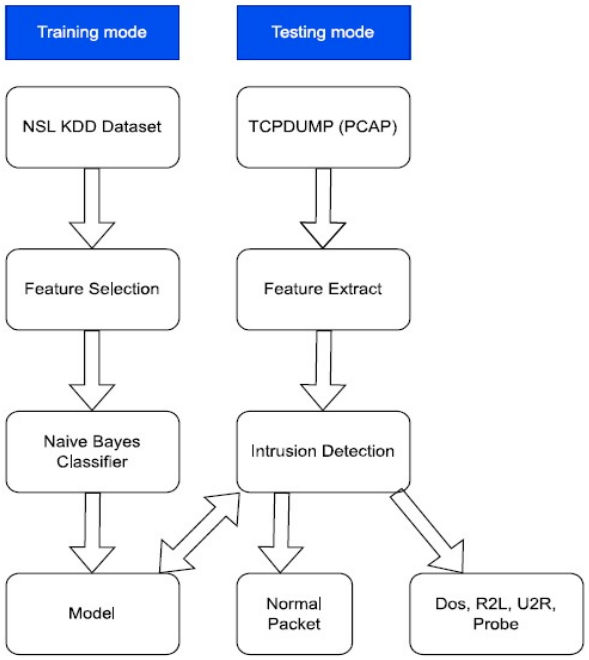


Fig 3. Architecture of proposed solution

A two-step intrusion detection procedure with training and testing modes is shown in Fig 3. A Naive Bayes classifier is used to create a predictive model after the NSL KDD dataset has undergone feature selection during the training phase. The model is used for intrusion detection after real-world network traffic recorded in PCAP format (using programs like TCPDUMP) is subjected to feature extraction during testing. Packets are classified by the system as normal or anomalous,

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and anomalies are further divided into DoS, R2L, U2R, and Probe attack types.

C. Detailed Module Implementation

1) Data Preprocessing and Model Training: The pipeline of network intrusion detection begins with preprocessing, in which the raw data is loaded into a Spark DataFrame. Feature engineering is then divided into two directions: categorical features are converted by StringIndexer and OneHotEncoder, and numerical features remain unchanged. These features are then merged into a concatenated feature vector through VectorAssembler. The pipeline continues to apply PySpark's MLlib Library to train three types of classification models: Random Forest, Logistic Regression, and Naive Bayes, with various strategies on network intrusion detection.

2) Feature Extraction: This module converts unstructured network traffic data into features that can be used for machine learning analysis. The module uses PyShark to parse PCAP files and methodically extract vital packet attributes, such as source and destination IPs, ports, protocols, packet sizes, timestamps, and TCP flags, starting with real-time packet captures from tcpdump. The module calculates connection-level aggregates like duration, total bytes transferred, packet counts, and flag ratios using these basic components. It also derives advanced metrics such as connection states, throughput measurements, inter-arrival times, and service type classifications to capture more complex network behaviors. The machine learning model receives a rich, structured representation of network activity patterns in the form of a finished feature set that has been arranged into a Pandas DataFrame and exported as a CSV file.

3) Real-time Detection: The real-time detection module operates by applying the previously trained ML model to identify network intrusions as they occur. The core classification process starts by deserializing the pre-trained model from its .sav file using the pickle module, then feeds the preprocessed network data through it to generate predictions. These predictions, whether in the form of class indices or one-hot encoded vectors, are ultimately translated back into meaningful categorizations such as Normal, DoS, or Probe.

D. Testing

During this stage, the model's accuracy and generalizability to new data were assessed using a different test set. By simulating a Denial-of-Service (DoS) attack with hping, the model was able to classify and react to anomalous network traffic based on its training, confirming its efficacy in identifying malicious activity.

IV. Results

The Table 4 depicts the performance of three models—Random Forest, Logistic Regression, and Naive Bayes—was evaluated in terms of accuracy. Among the models, Logistic Regression achieved an accuracy of 99.05%, closely followed by the Random Forest model with an accuracy of 98.72%. The Naive Bayes model attained an accuracy of 85.97%, and was ultimately chosen for its simplicity and generalizability, as the higher accuracy of the other models was likely a result of overfitting.

Fig 5, 6 and 7 depict the confusion matrices for the Random Forest, Logistic Regression and Naive Bayes models, showing the comparison between predicted and actual labels to assess model performance.

Model	Accuracy
Logistic Regression	99.05%
Random Forest	98.72%
Naive Bayes	85.97%

Table 1: Performance of three models

Random Forest Confusion Matrix

Actual ↓ / Predicted →	Normal	R2L	U2R	DoS	Probe
Normal	100	5	2	1	0
R2L	3	95	4	2	1
U2R	2	3	90	1	0
DoS	1	2	1	88	2
Probe	0	1	0	1	92

Fig 5. Confusion Matrix for Random Forest Classifier

Logistic Regression Confusion Matrix

Actual ↓ / Predicted →	Normal	R2L	U2R	DoS	Probe
Normal	95	7	3	2	1
R2L	5	90	5	3	2
U2R	3	4	85	2	1
DoS	2	3	2	83	3
Probe	1	2	1	2	87

Fig 6. Confusion Matrix for Logistic Regression Classifier

Actual ↓ / Predicted →	Normal	R2L	U2R	DoS	Probe
Normal	90	8	4	3	2
R2L	7	85	6	4	3
U2R	4	5	80	3	2
DoS	3	4	3	78	4
Probe	2	3	2	3	82

Fig 7. Confusion Matrix for Naive Bayes Classifier

V. Conclusion

Our real-time network intrusion detection system integrates data collection, feature engineering, and model training into one pipeline. TCPDump captures packets in real-time, and our feature engineering transforms raw network data into formatted inputs. We trained three classification models—Random Forest, Logistic Regression, and Naive Bayes—using PySpark to classify network traffic into five classes and selected the top-performing model to be the Naive Bayes classifier.

The second development phase will involve emulating R2L, U2R, and Probe attacks for testing the performance of

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our model on actual time data. Each subsequent version will further increase the detection accuracy, decrease false alarms, and make the system even more resilient to new network attacks.

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